

Management of patients with Gastrointestinal diseases

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الحديث الجانبي يشتت الانتباه



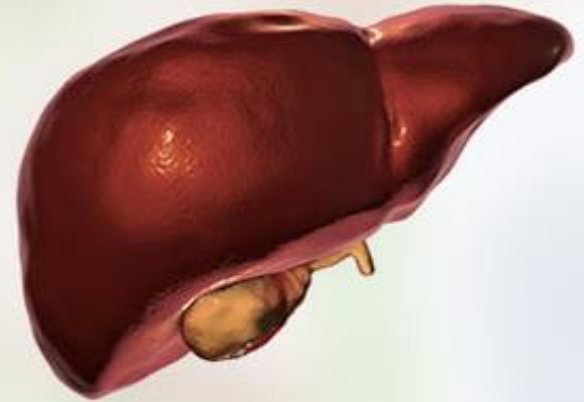
من فضلك اغلق الهاتف

4- liver cirrhosis & Esophageal varices

- ◉ Definition
- ◉ Causes
- ◉ Stages
- ◉ Manifestations
- ◉ Diagnosis
- ◉ Complications
- ◉ Managements

Definition of Liver Cirrhosis

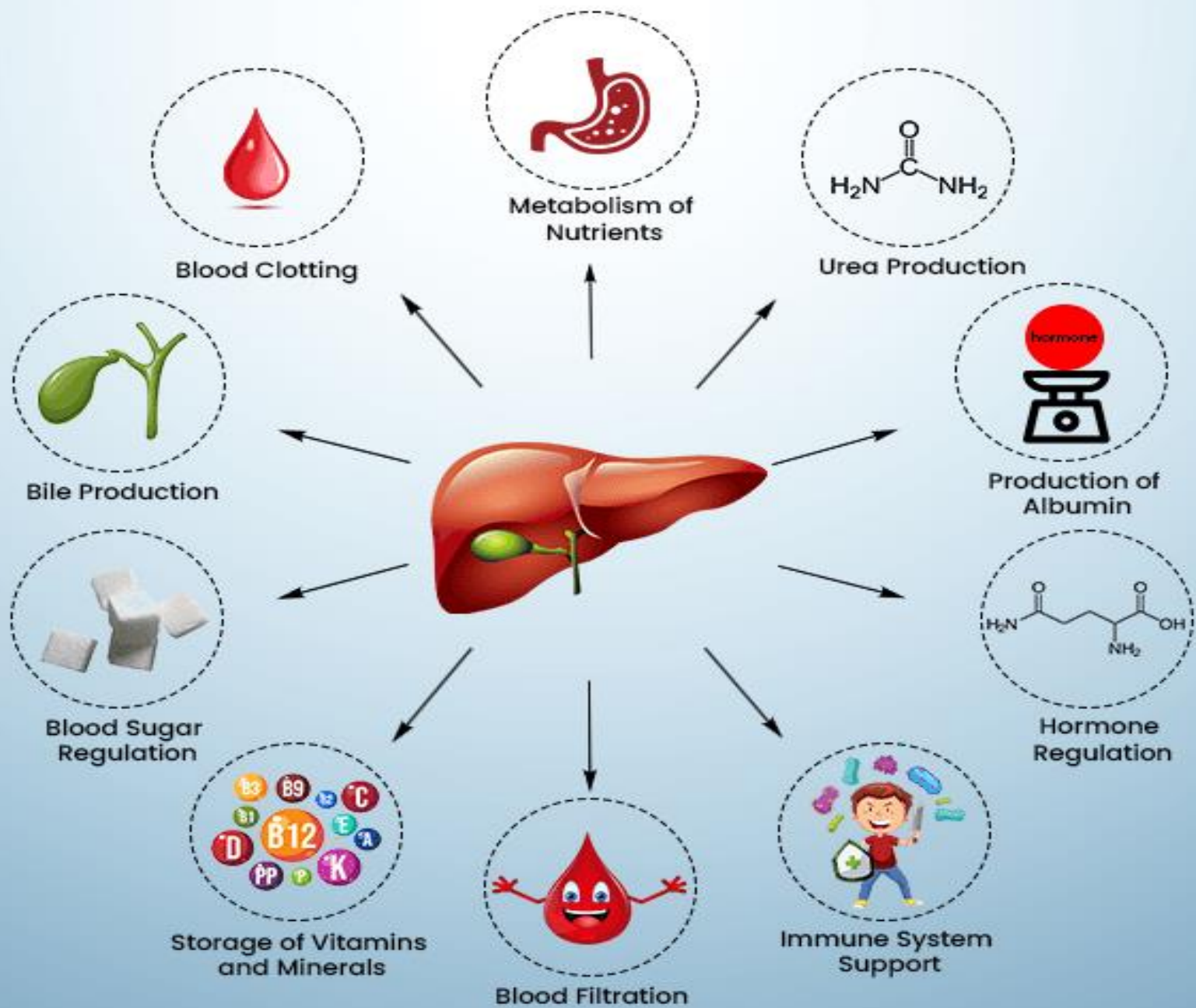
▶ **Chronic** progressive disease characterized by **fibrosis** (scarring) and **nodule formation** that replace normal liver tissue and **disrupts** the structure and **function** of the liver.



Healthy liver



Cirrhotic liver



The most common **Causes Of Cirrhosis** of the liver are:



Chronic Alcoholism



Chronic viral infections of the liver



Autoimmunity



Metabolic disorders



Fatty liver associated with
obesity and diabetes



Insufficient blood circulation in the
liver due to various causes



Stages of cirrhosis

Healthy liver



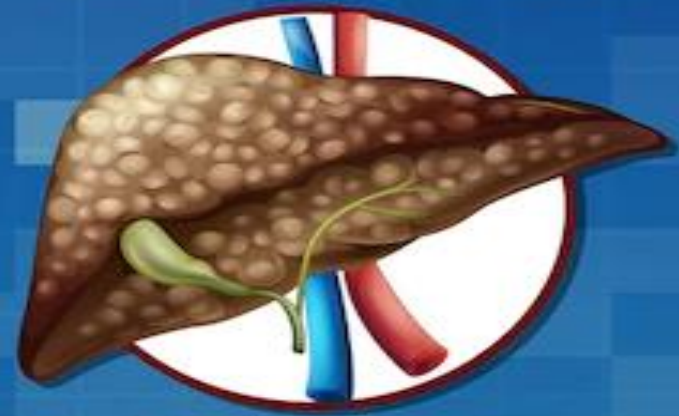
Fatty liver



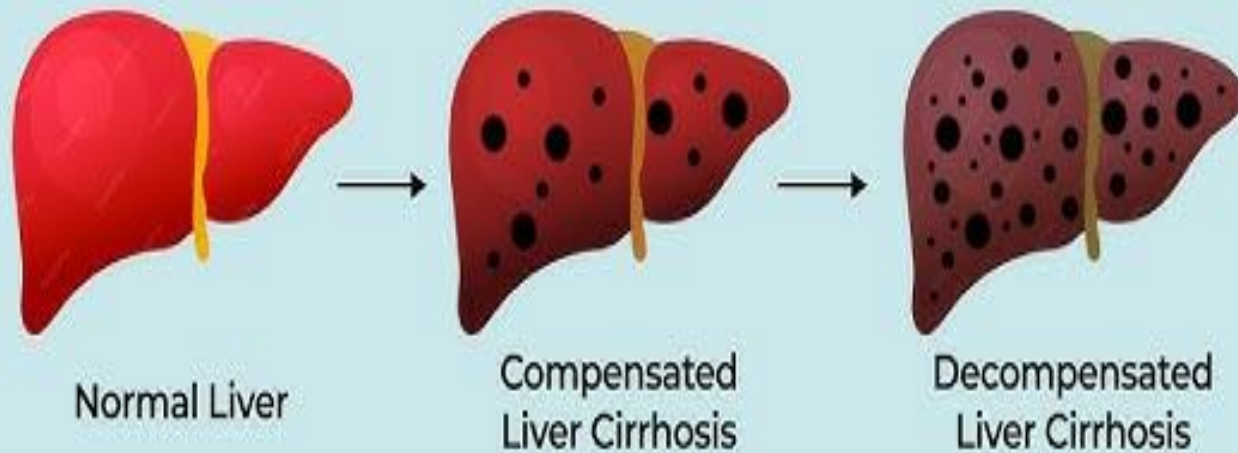
Fibrosis of liver



Cirrhosis of liver



Manifestation of liver cirrhosis



What is the difference.?

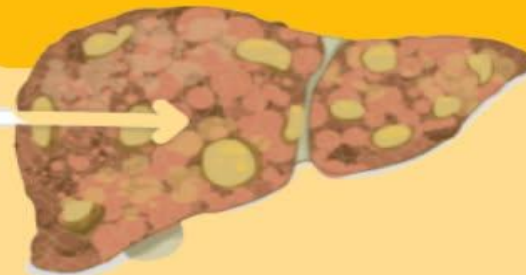
Compensated cirrhosis

- none or minor symptoms
- often goes unnoticed
- although liver is scarred it can still do its jobs



Decompensated cirrhosis

- develop complications of ascites, varices or confusion
- the liver can no longer do its job as before



- There are still enough healthy cells in the liver to do its function
- liver can **compensate** the previous damage
- Fatigue, anorexia, weight loss, **hepatomegaly** or splenomegaly.

Compensated

- **Failure** of the liver function to synthesize protein and clotting factors.

Decompensated

Manifestation of decompensated liver cirrhosis

Ascites



Jaundice



Weight loss



Epistaxis



Clubbing fingers



Fever



Assessment and diagnostic studies :

- ❖ **Presence of manifestation .**

- ❖ **lab investigations:**

Elevated Ast, Alt

Elevated serum ammonia level

Hyperbilirubinemia

Prolonged prothrombin time .

Low blood urea nitrogen level

- ❖ **Abdominal ultrasound** : enlarged liver.

- ❖ **CT, MRI** : give information about the liver size, blood flow and obstruction

- ❖ **Liver biopsy**



Complications of Liver Cirrhosis



Recurrent Infection



Altered behavior (Encephalopathy)



Swelling in the legs and abdomen.



Bleeding in vomiting and stool



Portal Hypertension



Infections



Jaundice



Malnutrition



Liver cancer



Consult with your **doctor** to avoid
these complications

Medical management

- ▶ **Interferon**: for management of viral hepatitis B and C.
- ▶ **corticosteroids** : for management of autoimmune hepatitis
- ▶ **Vitamin E** :to decrease damage in cirrhotic livers and reduce immune abnormalities that contribute to the development of the disease.
- ▶ **vitamin K**: for management of coagulation disorders
- ▶ **L-carnitine injections**: to improve circulation to the liver in people with cirrhosis

Nursing Interventions

1- Promoting rest

- Permit the liver to reestablish its functional ability, reduces the demands on the liver and increases the liver's blood supply
- **Fowler position** for maximal respiratory efficiency.
- **Oxygen therapy** to oxygenate the damaged cells and prevent further cell destruction

2- Improving Nutritional Status:

- The patient with cirrhosis who has **no ascites or edema** can receive a **high-protein** diet if tolerated, supplemented by **vitamins B complex ,A, C, K. and folic acid**).
- In advance stage of cirrhosis, patient should receive low protein, high-caloric diet.
- **Low** sodium diet
- **low** Fat diet
- **Frequent** meals are tolerated well than three large meals

3- Providing Skin Care:

- Frequently change of **position** to prevent pressure ulcers.
- **Avoid irritating** soaps and the use of adhesive tape to prevent trauma to the skin.
- Minimize **scratching** by the patient.

4- Reducing risk of injury:

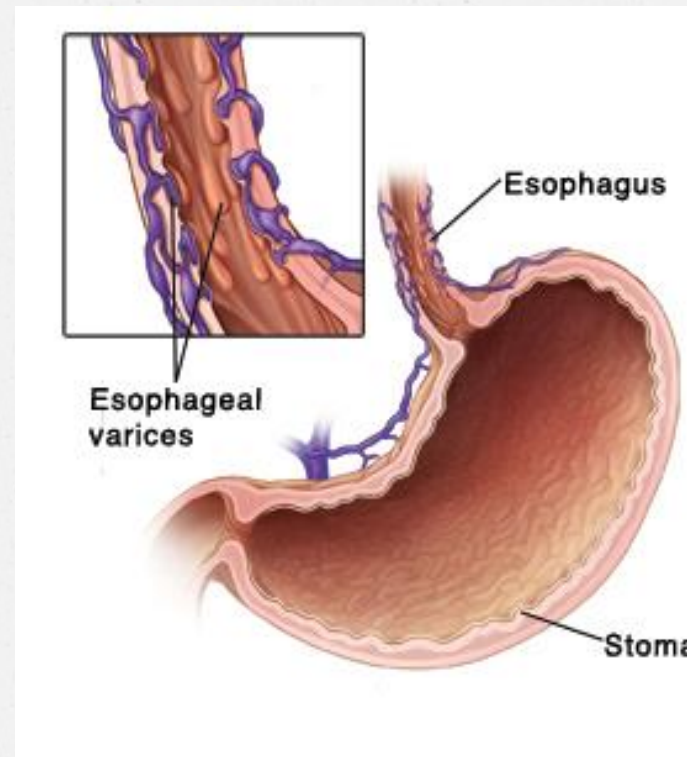
- Fall prevention
- Raised bed side rails
- A soft toothbrush
- Apply pressure to all venipuncture sites
- Remove sharp instruments from beside patient
- Frequent monitoring of vital signs and laboratory investigations

5– Teaching Patients Self-Care

- Adequate nutrition
- Period of rest
- Avoid hepatotoxic substances.
- Refer the client to the appropriate agency or support group for assistance with alcohol cessation
- Regular checkups and blood tests to follow the progress of the disease

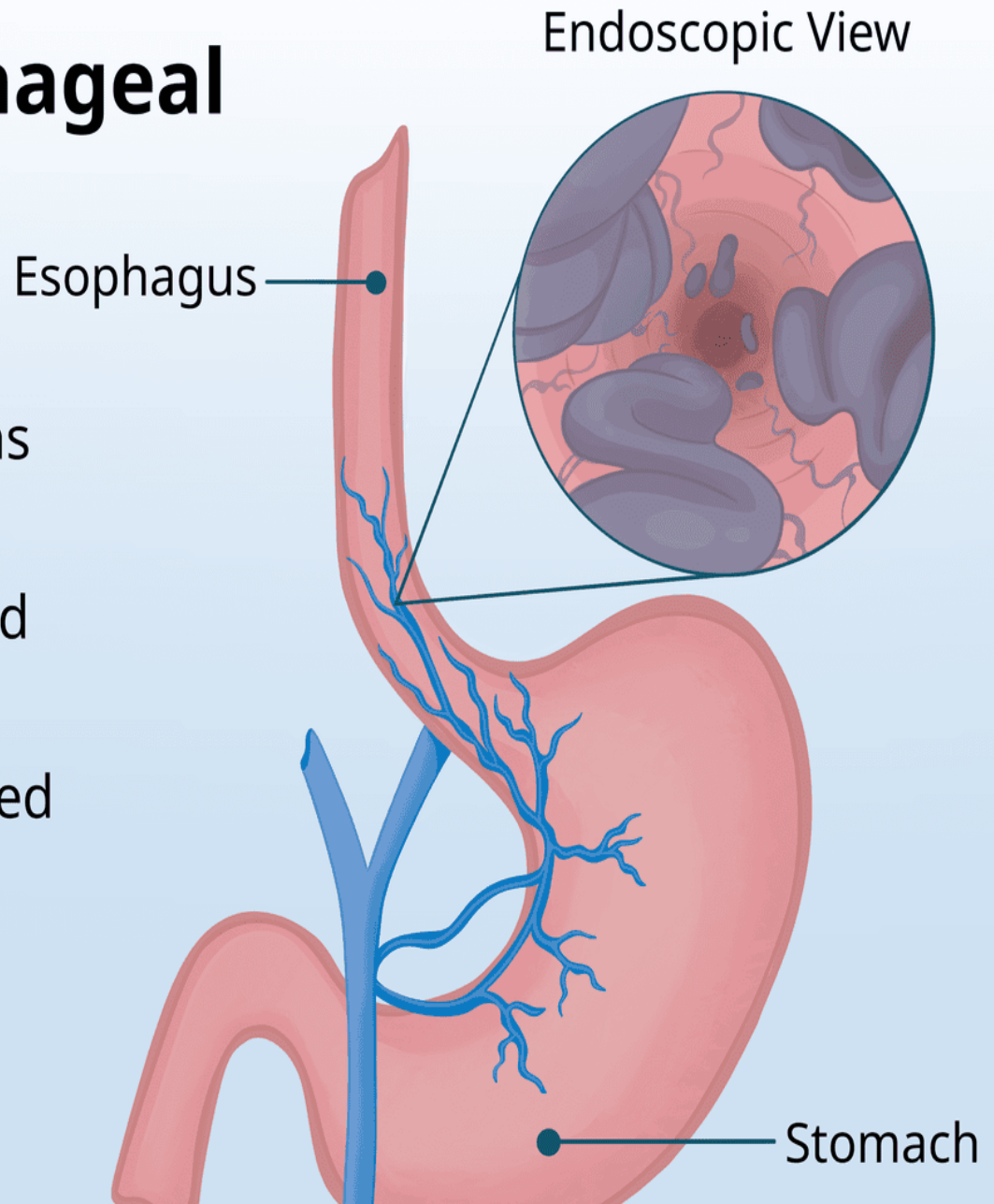
Esophageal varices are

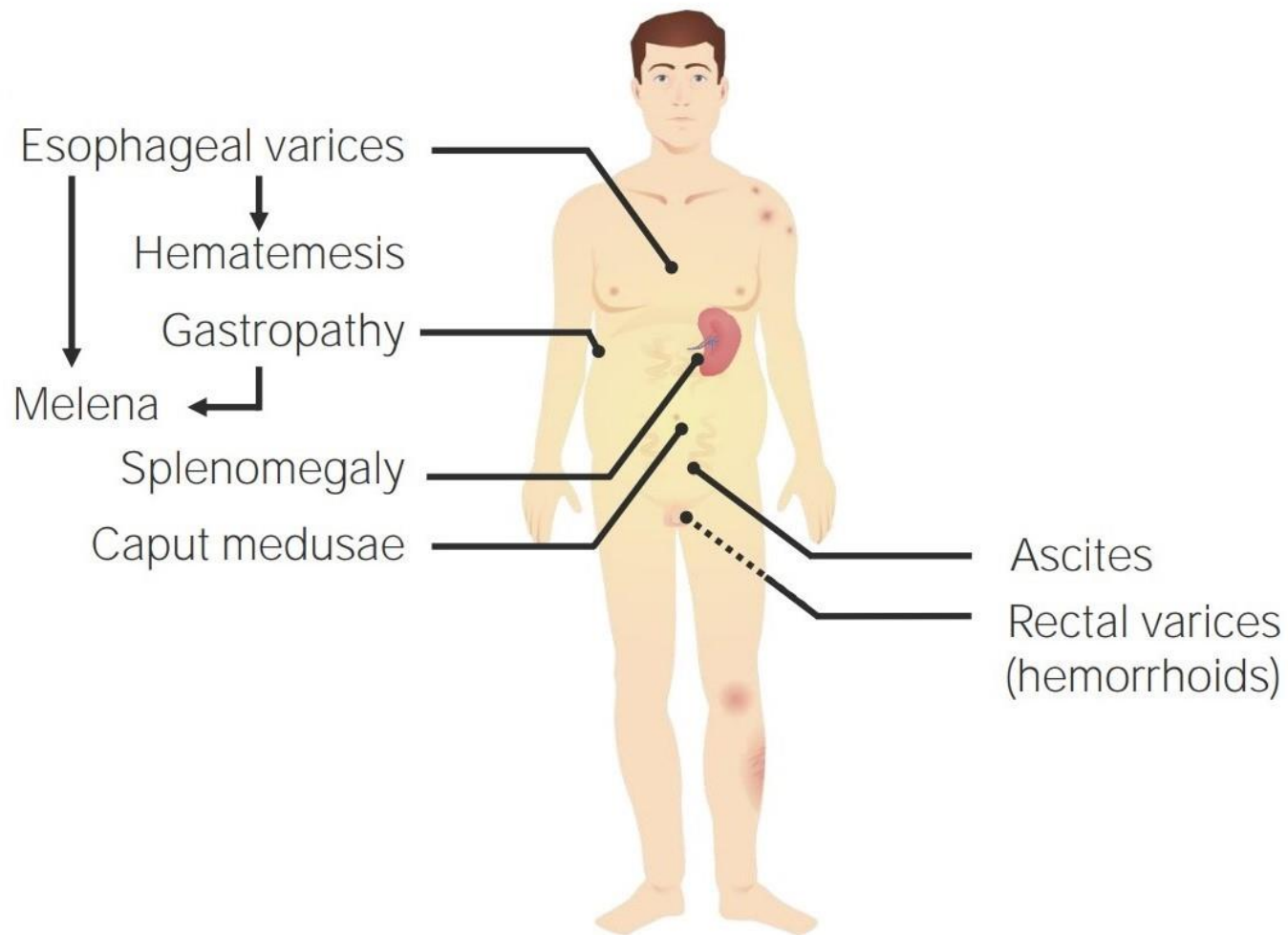
- Abnormally enlarged veins in the lower esophagus, usually caused liver cirrhosis and high blood pressure in the liver vein (portal hypertension)



What are Esophageal Varices?

Esophageal varices are swollen, thin-walled veins in the lower esophagus, the tube that carries food from the mouth to the stomach. Often associated with complications of cirrhosis.





Manifestation

- **Vomiting blood (hematemesis)**
- **Black, tarry stools (melena)**
- **Lightheadedness or shock (in severe bleeding)**



Diagnosis:

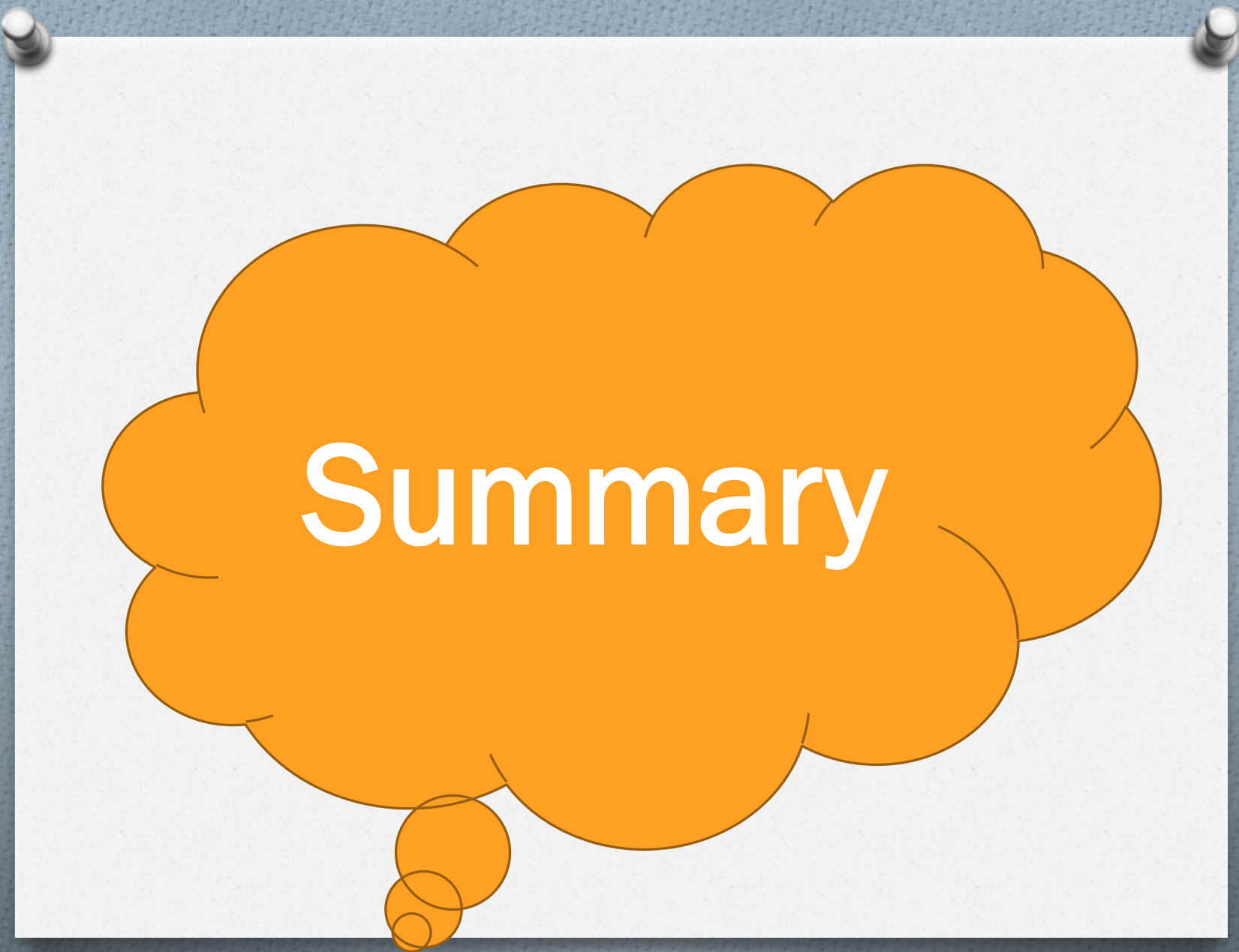
- Upper endoscopy (EGD) is the gold standard.
- Imaging like ultrasound with Doppler, CT, or MRI may also show portal hypertension.
- Blood tests may show liver dysfunction (e.g., high bilirubin, low platelets).



Nursing interventions

- Monitor vital signs every 15–30 minutes
- Maintain NPO status until cleared
- Establish large-bore IV access for fluids/blood transfusion
- Administer blood products as ordered
- Position patient in high Fowler's or side-lying to reduce aspiration risk

- Administer medications:
 - vasopressin to reduce portal pressure
 - Lactulose if hepatic encephalopathy is present
- Prepare for emergency endoscopy
- Monitor for signs of hypovolemic shock: ↓ BP, ↑ HR, cold/clammy skin



Summary

- List 3 complication for liver cirrhosis
- List 4 nursing interventions for liver cirrhosis
- List 4 nursing interventions for esophageal varices





Thank
you